START HERE — Read This First

What is this project?

An online shop (like a tiny Amazon) built the way real companies build software:

6 small Java programs ("microservices") that work together, deployed with the same tools professionals use — Docker, Kubernetes, AWS (simulated), and automated pipelines.

The 6 services, in one line each

Service	Job	Port
user-service	Register, login, give out JWT tokens	8081
product-service	Show products (with a Redis cache for speed)	8082
order-service	Take orders, run the "order saga"	8083
inventory-service	Track stock, reserve it when someone orders	8084
payment-service	Charge the customer (simulated)	8085
notification-service	Send emails when things happen	8086

Dictionary — every scary word, in plain English

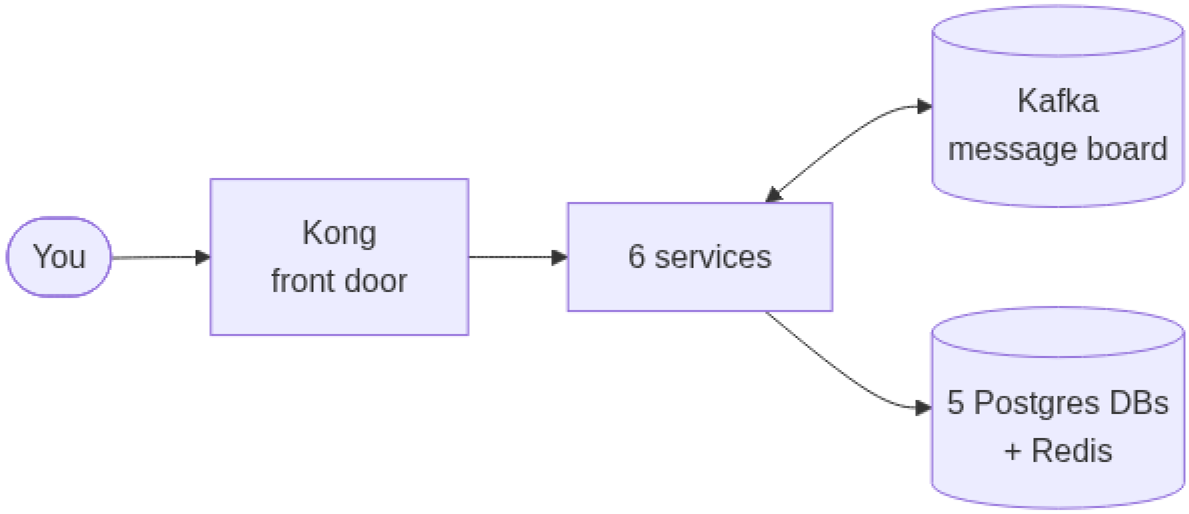
Word	Meaning
Container	One program packed with everything it needs, running in isolation. Like a food truck: kitchen + ingredients + menu, all in one box.
Docker	The tool that builds and runs containers.
Image	The frozen recipe/blueprint of a container. `docker build` makes an image; running it makes a container.
Pod	Kubernetes' word for "one running container that I manage."
Kubernetes (K8s)	The head office that manages hundreds of containers: restarts dead ones, adds more when busy. It never cooks — it only manages trucks.

Deployment	An order to Kubernetes: "keep N copies of this app alive, always." For apps that hold no data.
StatefulSet	Same, but for apps that DO hold data (databases). Each gets a permanent disk.
Service (K8s)	A fixed internal phone number. Callers dial the name; K8s connects them to any healthy pod.
Kafka	A message board between services. One service posts an event, others read it. Services never call each other directly.
Redis	A super-fast memory store. Used here for: login sessions, product cache, stock locks.
Kong	The API gateway — the single front door. Every request enters here first and gets routed to the right service.
AWS	Amazon's cloud — rent servers/databases instead of owning them.
Floci	A fake AWS running on your laptop. Same commands, zero cost.
CI/CD	Robots that test and deploy your code automatically every time you push to GitHub.
JWT	A signed ticket proving who you are. Login once, show the ticket on every request.

The learning order (do NOT skip ahead)

```
1. 01-DOCKER.md        → how one service becomes a container
2. 02-KUBERNETES.md    → how containers are managed at scale
3. 03-KAFKA.md         → how services talk without calling each other
4. 04-AWS-FLOCI.md     → what the cloud pieces are for
5. 05-CICD.md          → how code goes from git push to running in production
6. 06-RUNBOOK.md       → actually run everything + debug commands
```

The 30-second big picture



A request enters through Kong, hits one service, that service saves to its own

database and posts events on Kafka; other services react to those events. That's the whole system. Every doc after this just zooms into one part.